

AP Police SI Prelims 2026 — Quick-Revision Sheet

Last-minute condensed sheet: every key Arithmetic formula plus high-frequency GS facts.

ARITHMETIC FORMULAS

Percentage: $x\% = x/100$. Increase by $x\% \rightarrow x(1+x/100)$. Decrease $\rightarrow x(1-x/100)$. Rise $x\%$ then fall $x\% \rightarrow \text{net } -(x^2/100)\%$. Spend-same after price rise $x\% \rightarrow \text{cut consumption by } x/(100+x) \times 100\%$.

Profit & Loss: $\text{Profit}\% = (SP-CP)/CP \times 100$. $SP = CP \times (100+P\%)/100$. Two discounts $x\%, y\% \rightarrow x+y-xy/100$.

Interest: $SI = PRT/100$. CI: $A = P(1+R/100)^T$; $CI = A-P$. CI-SI (2 yrs) $= P(R/100)^2$.

Ratio & Average: Divide A in $m:n \rightarrow A \cdot m/(m+n), A \cdot n/(m+n)$. Average = sum/count. Avg speed (equal distance, speeds a, b) $= 2ab/(a+b)$.

Time & Work: 1 day's work $= 1/n$. Together $= AB/(A+B)$. Efficiency $\propto 1/\text{time}$.

Speed & Distance: Speed = dist/time. $1 \text{ km/h} = 5/18 \text{ m/s}$. Train + pole: $t = L/\text{speed}$. Train + platform: $t = (L+P)/\text{speed}$. Downstream = boat+stream; upstream = boat-stream.

Mensuration: Rectangle area $l \times b$, perimeter $2(l+b)$. Square $a^2, 4a$. Triangle $\frac{1}{2} \cdot b \cdot h$. Circle $\pi r^2, 2\pi r$. Cuboid $l \cdot b \cdot h$. Cube $a^3, 6a^2$. Cylinder $\pi r^2 h, \text{CSA } 2\pi r h$. Sphere $(4/3)\pi r^3, 4\pi r^2$. Cone $(1/3)\pi r^2 h$.

Numbers: $\text{LCM} \times \text{HCF} = \text{product of two numbers}$. Sum of first $n = n(n+1)/2$; squares $n(n+1)(2n+1)/6$; cubes $[n(n+1)/2]^2$. Divisibility: 3 (digit-sum), 9 (digit-sum), 11 (alt-sum).

Useful fractions: $1/2=50\% \cdot 1/3 \approx 33.3\% \cdot 1/4=25\% \cdot 1/5=20\% \cdot 1/6 \approx 16.7\% \cdot 1/8=12.5\% \cdot 3/8=37.5\%$.

REASONING QUICK TIPS

- Write A=1 ... Z=26 before coding questions.
- Family-tree symbols for blood relations; compass sketch for direction.
- Series: check +constant, $\times$ constant, squares/cubes, alternating, prime, Fibonacci.
- Dice/cube: opposite faces never show together.
- Mirror image: left $\leftrightarrow$ right flips; water image: top $\leftrightarrow$ bottom flips.

POLITY KEY FACTS

- Constitution effective **26 Jan 1950**; drafting chaired by **B. R. Ambedkar**.
- Fundamental Rights — Part III; Directive Principles — Part IV; **11** Fundamental Duties.
- Parliament = Lok Sabha (lower) + Rajya Sabha (upper). Money bills only in Lok Sabha.
- President elected by an electoral college; PM heads the government.
- Panchayati Raj = **3 tiers**. Election Commission & UPSC are constitutional bodies.

HISTORY KEY FACTS

- Indus Valley ~2500 BCE (town planning). Ashoka — **Maurya**; Guptas — "Golden Age."
- Battle of Plassey **1757**; Revolt of **1857** (began at Meerut).
- INC founded **1885**; Quit India **1942** ("Do or Die" — Gandhi); Independence **1947**.
- Vijayanagara — South India. **Potti Sriramulu** → formation of Andhra State (1953).

GEOGRAPHY KEY FACTS

- Longest river: **Ganga**. SW monsoon arrives ~**June**. IST based on **82.5°E**.
- AP rivers: **Godavari & Krishna**. Major AP port: **Visakhapatnam**. Capital: **Amaravati**.
- Himalayas = fold mountains. Deccan = basalt. Kharif (monsoon) / Rabi (winter) crops.

GENERAL SCIENCE KEY FACTS

- Force unit **newton**; current **ampere**; light $\approx 3 \times 10^8$ m/s.
- Neutral pH **7**. Salt = **NaCl**; water = H₂O; respiration uses **oxygen**.
- Cell powerhouse = **mitochondrion**. Bones in adult = **206**. Largest gland = **liver**.
- Vit C → scurvy; Vit D → from sunlight; Malaria → protozoan. Diamond = hardest.
- Sound can't travel in **vacuum**. Photosynthesis in **leaves** (releases O₂).

ECONOMY KEY FACTS

- **RBI** = central bank (monetary policy, issues currency). **NITI Aayog** replaced Planning Commission.
- GDP = total goods & services. Inflation = general price rise. Financial year **Apr-Mar**.
- Union Budget by the **Finance Minister**; Economic Survey precedes the Budget.

CURRENT AFFAIRS SNAPSHOT (2026 — verify before exam)

- President of India: **Droupadi Murmu**. Union Budget 2026-27 by **Nirmala Sitharaman**.
- AP Chief Minister: **N. Chandrababu Naidu**. **Amaravati** confirmed as permanent capital (2026).
- AP vision: **Swarna Andhra 2047**; "**Super Six**" welfare scheme promises.
- India hosting the **18th BRICS Summit (2026)**; hosted the **India-AI Impact Summit** (New Delhi); attended **G7 2026** (France).
- *Current affairs change fast — confirm the latest before the exam.*

NATIONAL SYMBOLS / GK

National animal **tiger** · national bird **peacock** · capital **New Delhi** · currency **rupee** (issued by RBI) · ISRO HQ **Bengaluru** · Telangana split from AP in **2014** · India's longest land border with **Bangladesh**.

Condensed for revision. Cross-check the syllabus and all current facts against the official APSLPBRB notification and up-to-date sources.